

| Paper                           | Modello                                                                                               | Task      | Dataset (Split)                                                                          | Orizzonte $\tau$                                       | Osservazione                               | Input                                                                     | Architettura                         | Task ausiliari                                              |
|---------------------------------|-------------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------|---------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------------------|
| Koppula et al. (2015)           | ATCRF (Anticipatory Temporal Conditional Random Field)                                                | STA       | CAD-120                                                                                  | 1-10s                                                  | 0-30s                                      | RGB                                                                       | Graph-based                          | Predizione affordance oggetti                               |
| Vondrick et al. (2016)          | AVR                                                                                                   | STA       | TV Human Interaction                                                                     | 1s                                                     | 0s                                         | RGB                                                                       | CNN                                  | Frame forecasting                                           |
| Furnari et al. (2018)           | VNMCE (Verb-Noun Marginal Cross Entropy)                                                              | STA       | Epic-Kitchens 55 (Val)                                                                   | 1s                                                     | 1s                                         | RGB / Past Actions Segments                                               | CNN                                  | Action recognition                                          |
| Yeung et al. (2018)             | Multi-LSTM                                                                                            | STA       | MultiTHUMOS                                                                              | 0-2s                                                   | 1.5s                                       | RGB                                                                       | RNN/LSTM                             | Action segmentation                                         |
| Furnari & Farinella (2019/2020) | RU-LSTM                                                                                               | FIA       | Epic Kitchens 55 (S1, S2, Val) / EGTEA Gaze+                                             | 0.25–2s                                                | 1.75-3.5 s                                 | RGB / Flow/ Object Detections                                             | RNN/LSTM                             | Action recognition                                          |
| Ke et al. (2019)                | TOS_AF_TSC (Time-conditioned One-Shot + Attended Temporal Feature + Time-conditioned Skip Connection) | STA       | 50 Salads / Epic Kitchens 55 (S1)                                                        | 1-60s                                                  | 30s                                        | RGB                                                                       | RNN/LSTM                             | Action recognition                                          |
| Miech et al. (2019)             | Predictive + Transitional (due varianti: AR e VA)                                                     | STA       | ActivityNet / Breakfast Actions / Epic Kitchen 55 (S2, Val)                              | 1s                                                     | Sconosciuto                                | RGB                                                                       | CNN                                  | Nessuno                                                     |
| Guan et al. (2020)              | AGG                                                                                                   | STA       | Epic Kitchens 55                                                                         | 1-45s                                                  | 1-2m                                       | RGB / Past Actions Segments                                               | Generative Model                     | Nessuno                                                     |
| Liu M. et al. (2020)            | FHOI                                                                                                  | STA       | Epic Kitchens 55 (S1, S2) / EGTEA Gaze+                                                  | 0.5-1s                                                 | 2.7s                                       | RGB / Object Detections                                                   | CNN                                  | Hand motion forecasting, Interaction hotspot prediction     |
| Nagarajan et al. (2020)         | Ego-Topo                                                                                              | STA       | Epic Kitchens 55 (Val) / EGTEA Gaze+                                                     | 0.5s-1s                                                | 1.5s-40m                                   | RGB                                                                       | Graph-based                          | Nessuno                                                     |
| Rasouli et al. (2020)           | SF-GRU (Stacked Feature GRU)                                                                          | STA       | Pedestrian Intention Estimation                                                          | 2s                                                     | 0.5s                                       | RGB / Detected People / Poses / Proprioceptive Data/ Past Action Segments | RNN/GRU                              | Nessuno                                                     |
| Sener et al. (2020/2021)        | TempAgg                                                                                               | STA / LTA | 50 Salads / Breakfast Actions / Epic-Kitchens 55 (S1, S2, Val)                           | 1s                                                     | 1-2m                                       | RGB / Flow / Object Detections / Past Action Segments                     | Attention-based Temporal Aggregation | Nessuno                                                     |
| Wu Y. et al. (2020)             | ImagineRNN                                                                                            | STA       | Epic Kitchens 55 (S1, S2, Val) / EGTEA Gaze+                                             | 0.25-1.5s                                              | 2.5s                                       | RGB / Flow / Object Detections                                            | LSTM                                 | Nessuno                                                     |
| Zhang T. et al. (2020)          | IAI (Intuition And Analysis Integration)                                                              | STA       | Epic Kitchens 55 (S2, Val)                                                               | 1s                                                     | 11s                                        | RGB / Flow / Object Detections                                            | LSTM                                 | Action segmentation                                         |
| Camporesse et al. (2021)        | RU-LSTM con KD                                                                                        | STA       | Epic Kitchens 55 (S1, S2, Val) / EGTEA Gaze+                                             | 0.5s (EGTEA Gaze+) / 1s (EK55)                         | 1.75-3.5s                                  | RGB / Flow / Object Detections                                            | LSTM                                 | Nessuno                                                     |
| Dessalene et al. (2021)         | Ego-OMG (Egocentric Object Manipulation Graphs)                                                       | STA       | Epic Kitchens 55 (S1, S2)                                                                | 1s                                                     | 60s                                        | RGB                                                                       | LSTM                                 | Hand-object interaction mask prediction                     |
| Fernando & Herath (2021)        | ---                                                                                                   | STA       | Breakfast Actions / Epic Kitchens 55 (Val) / JHMDB / UCFL101                             | 1s                                                     | 2s                                         | RGB / Flow / Object Detections                                            | CNN                                  | Nessuno                                                     |
| Girdhar & Grauman (2021)        | AVT (Anticipative Video Transformer)                                                                  | STA       | 50 Salads / Epic Kitchens 55 (Val) / Epic Kitchens 100 (Val, Test) / EGTEA Gaze+         | 0.5-1s                                                 | Sconosciuto                                | RGB / Object Detections                                                   | Transformer                          | Frame forecasting, Action segmentation                      |
| Mahmud et al. (2021)            | ---                                                                                                   | STA / LTA | Epic-Kitchens 55 (Val) / MPII-Cooking / MPII-Cooking 2 / VIRAT Ground                    | 1s (EK55)                                              | 3 attività osservate precedenti (features) | RGB / Flow / Object Detections                                            | LSTM                                 | Video captioning delle azioni future (encoder-decoder LSTM) |
| Osman et al. (2021)             | SF-RU-LSTM (SlowFast Rolling-Unrolling LSTM)                                                          | STA       | Epic Kitchens 55 (Val) / EGTEA Gaze+                                                     | 0.5-2s                                                 | 3s                                         | RGB / Flow / Object Detections                                            | LSTM                                 | Nessuno                                                     |
| Qi et al. (2021)                | SRL (Self-Regulated Learning)                                                                         | FIA       | 50 Salads / Breakfast Actions / Epic-Kitchens 55 (S1, S2, Val) / EGTEA Gaze+             | 1s (EK55) / 0.5s (EGTEA Gaze+)                         | 1.5s                                       | RGB / Flow / Object Detections                                            | RNN/GRU                              | Temporal Contrastive Learning (TC)                          |
| Roy & Fernando (2021)           | MM-Transformer (Multi-modal transformer)                                                              | STA / LTA | 50Salads / Breakfast Actions / Epic-Kitchens 55 (S1, S2, Val)                            | 1s-3m / 1s (EK55)                                      | 1-5s                                       | RGB / Detected People / Object Detections                                 | Transformer                          | Nessuno                                                     |
| Xu M. et al. (2021)             | LSTR (Long Short-Term TRansformer)                                                                    | STA       | THUMOS'14 (Avg) / TVSeries (Avg)                                                         | 2s                                                     | 0-1024s                                    | RGB / Flow                                                                | Transformer                          | Action recognition                                          |
| Zatsarynna et al. (2021)        | MM-TCN                                                                                                | STA       | Epic Kitchens 55 (S1, S2) / Epic Kitchens 100 (Test)                                     | 1s                                                     | 5.25s                                      | RGB / Flow / Object Detections                                            | TCN                                  | Nessuno                                                     |
| Gupta et al. (2022)             | A-ACT                                                                                                 | STA / LTA | 50 Salads / Breakfast Actions / Epic-Kitchens 55 (S1, S2, Val)                           | 1s                                                     | Sconosciuto                                | RGB / Flow / Object Detections                                            | Transformer                          | Nessuno                                                     |
| Liu S. et al. (2022)            | OCT                                                                                                   | ---       | Epic-Kitchens 55 (Val) / Epic-Kitchens 100 (Val)                                         | 1s                                                     | Sconosciuto                                | RGB                                                                       | Transformer                          | Nessuno                                                     |
| Liu T. et al. (2022)            | HRO                                                                                                   | FIA       | Epic-Kitchens 55 (Val) / EGTEA Gaze+                                                     | 0.25-2s                                                | 1.5-2s                                     | RGB / Flow / Object Detections                                            | RNN/GRU                              | Nessuno                                                     |
| Nawhal et al. (2022)            | ANTICIPATR                                                                                            | STA / LTA | 50 Salads / Breakfast Actions / Epic-Kitchens 55 (Val) / EGTEA Gaze+                     | 1s (EK55)                                              | Sconosciuto                                | RGB                                                                       | Transformer                          | Nessuno                                                     |
| Roy & Fernando (2022/2024)      | Abstract Goal / Latent Goal                                                                           | NAT       | Epic Kitchens 55 (S1, S2, Val) / Epic Kitchens 100 (Test) / EGTEA Gaze+                  | 1s                                                     | 2-15s                                      | RGB / Flow / Object Detections                                            | RNN/GRU                              | Nessuno                                                     |
| Wu C. et al. (2022)             | MeMVIT (Memory-augmented Multiscale Vision Transformer)                                               | STA       | Atomic Visual Actions (AVA) / Epic Kitchens 100 (Val)                                    | 1s                                                     | 70s                                        | RGB                                                                       | Transformer                          | Nessuno                                                     |
| Xu X. et al. (2022)             | DCR (Dynamic Context Removal)                                                                         | STA       | 50 Salads / Epic Kitchens 55 (S1, S2, Val) / Epic Kitchens 100 (Val, Test) / EGTEA Gaze+ | 0.25-1s                                                | 5-10s                                      | RGB / Flow / Object Detections                                            | Transformer                          | Frame forecasting, Frame order recovery                     |
| Girase et al. (2023)            | RAFTformer                                                                                            | STA       | Epic Kitchens 55 / Epic Kitchens 100 (Val, Test) / EGTEA Gaze+                           | 1s                                                     | Sconosciuto                                | RGB                                                                       | Transformer                          | Nessuno                                                     |
| Manousaki et al. (2023)         | VLMHA (Visual-Linguistic Modeling of Action History)                                                  | STA       | 50 Salads / Assembly101 / Meccano                                                        | 0.25-2s                                                | Sconosciuto                                | RGB / Gaze / Hand Masks / Past Action Segments                            | RNN                                  | Nessuno                                                     |
| Wang et al. (2023)              | MAT (Kinetics Pretraining)                                                                            | STA       | HDD / Epic Kitchens 100 (Val) / THUMOS / TV Series                                       | 1s                                                     | Sconosciuto                                | RGB / Flow / Object Detections                                            | Transformer                          | Action recognition                                          |
| Zhong et al. (2023)             | AFFT (Anticipative Feature Fusion Transformer                                                         | FIA       | Epic-Kitchens 100 (Val, Test) / EGTEA Gaze+                                              | 0.5s                                                   | 4-20s                                      | RGB / Flow / Object Detections / Audio                                    | Transformer                          | Action recognition                                          |
| Diko et al. (2024)              | S-GEAR (Semantically Guided Representation Learning)                                                  | STA       | 50 Salads / Epic Kitchens 55 (Val) / Epic Kitchens 100 (Val, Test) / EGTEA Gaze+         | 0.5-1s                                                 | 10s (EK55, EGTEA Gaze+), 15s (EK100)       | RGB / Flow / Object Detections                                            | Transformer                          | Nessuno                                                     |
| Guermal et al. (2024)           | JOADAA (Joint Online Action Detection and Action Anticipation)                                        | STA       | Charades / THUMOS / MultiTHUMOS                                                          | 0.03-0.2s                                              | Sconosciuto                                | RGB                                                                       | Transformer                          | Online action detection                                     |
| Guo et al. (2024)               | UADT (Uncertainty-Aware Action Decoupling Transformer)                                                | STA       | 50 Salads / Epic Kitchens 100 (Val) / EGTEA Gaze+                                        | 0.5-1s                                                 | 2-3s                                       | RGB / Flow / Object Detections                                            | Transformer                          | Action recognition                                          |
| Mittal et al. (2024)            | PlausiVL                                                                                              | STA       | Ego4D / Epic Kitchens 100 (Val)                                                          | 0.5s-3m                                                | Sconosciuto                                | RGB / Past Actions Segments                                               | Transformer                          | Nessuno                                                     |
| Roy et al. (2024)               | InAVIT                                                                                                | STA       | Epic Kitchens 100 (Val, Test) / EGTEA Gaze+                                              | 1s (EK100) / 0.5s (EGTEA)                              | 2s                                         | RGB / Object Detections / Hand Masks                                      | Transformer                          | Nessuno                                                     |
| Thakur et al. (2024)            | NAOGAT (Next-Active-Object Guided Anticipation Transformer)                                           | NAT       | Ego4D / Epic Kitchens 100 (Val)                                                          | TTC (time-to-contact) variabile (predetto come output) | 4s                                         | RGB / Object detections                                                   | Transformer                          | Next active object prediction                               |
| Zhang Y. et al. (2024)          | SEAD                                                                                                  | STA       | CAD-120, Charades, Home Action Genome                                                    | 1s                                                     | ---                                        | RGB                                                                       | Graph-based                          | Nessuno                                                     |